

Organización, arquitectura y plataformas de sistemas embebidos

Objetivo:

El alumno identificará las arquitecturas actuales de sistemas embebidos con plataformas comerciales.

Objetivo:

El alumno aprenderá los fundamentos de una arquitectura de computadoras...

...digo, de un sistema embebido... sí, eso...

Temario:

1. Procesador
2. Mapa de memoria del sistema
3. Controlador de interrupciones
4. Temporizadores
5. Memoria volátil y no volátil. Direccinamientos
6. Bus serie universal
7. Interconexión de dispositivos
8. Secuencia de arranque en plataformas embebidas
9. Estudio de plataformas comerciales para sistemas embebidos (INTEL, ARM, etc.)

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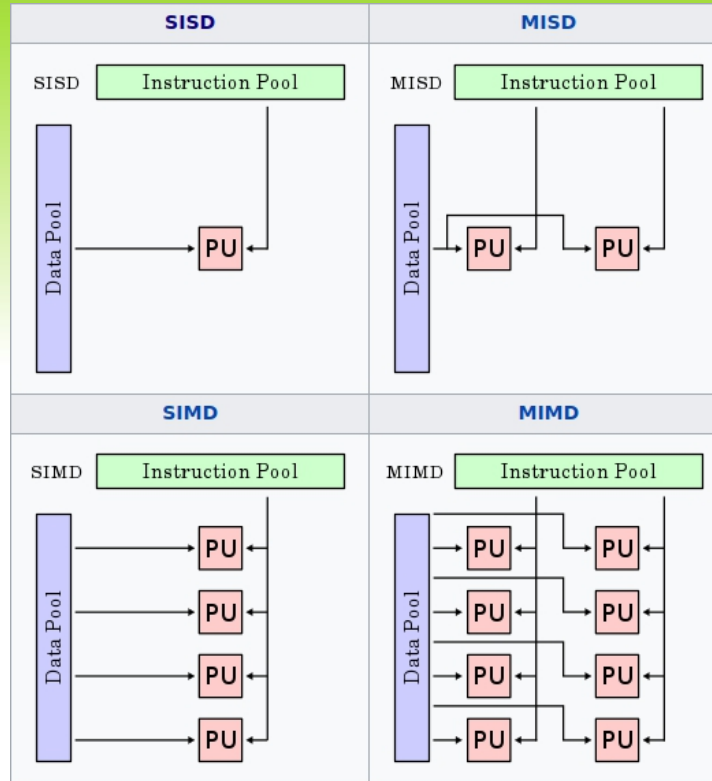
0. Tipos de computadora

(Arquitectura)

Tipos de computadoras (arquitectura CPU):

- SISD
- SIMD
- MISD
- MIMD

Tipos de computadoras:



1. Procesador

Procesador:

Se puede definir conceptualmente a una computadora como una máquina que consta de elementos de entrada, elementos de salida, un procesador central y una memoria

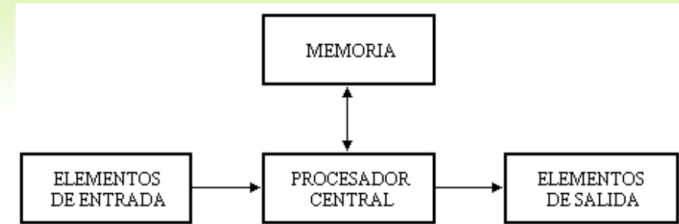


Figura 1.1. Diagrama de bloques de una computadora.

Procesador:

Definición: Elemento central de una computadora encargado realizar cómputo con la información (o datos) contenida en la memoria principal mediante la ejecución del programas contenido también en dicha memoria.

Procesador:

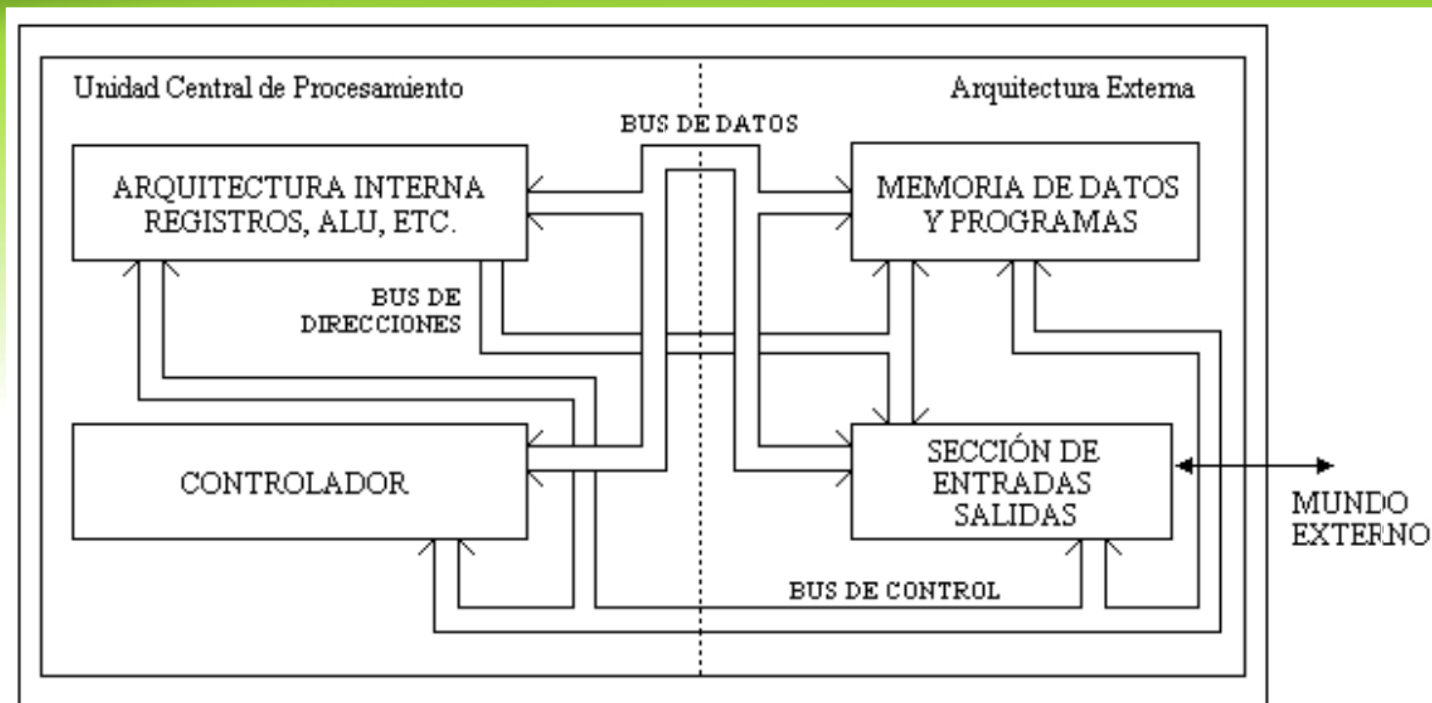


Figura 1.2. Computadora de von Neumann.

Procesador:

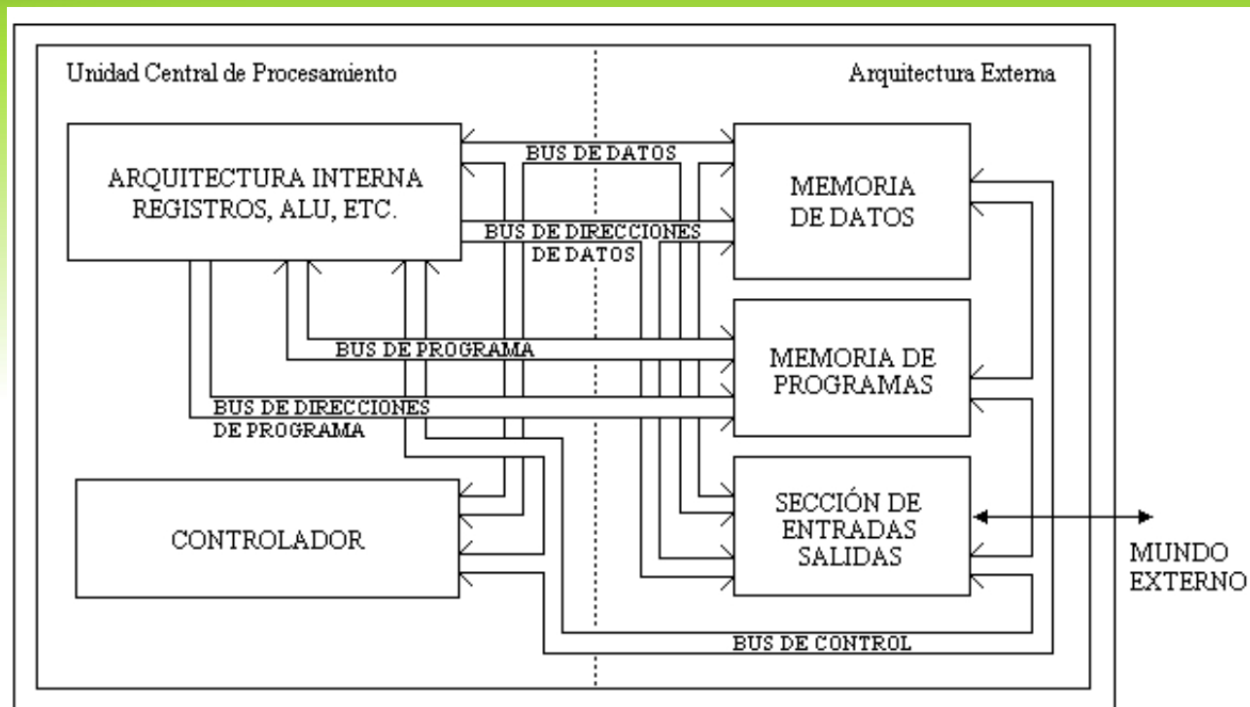
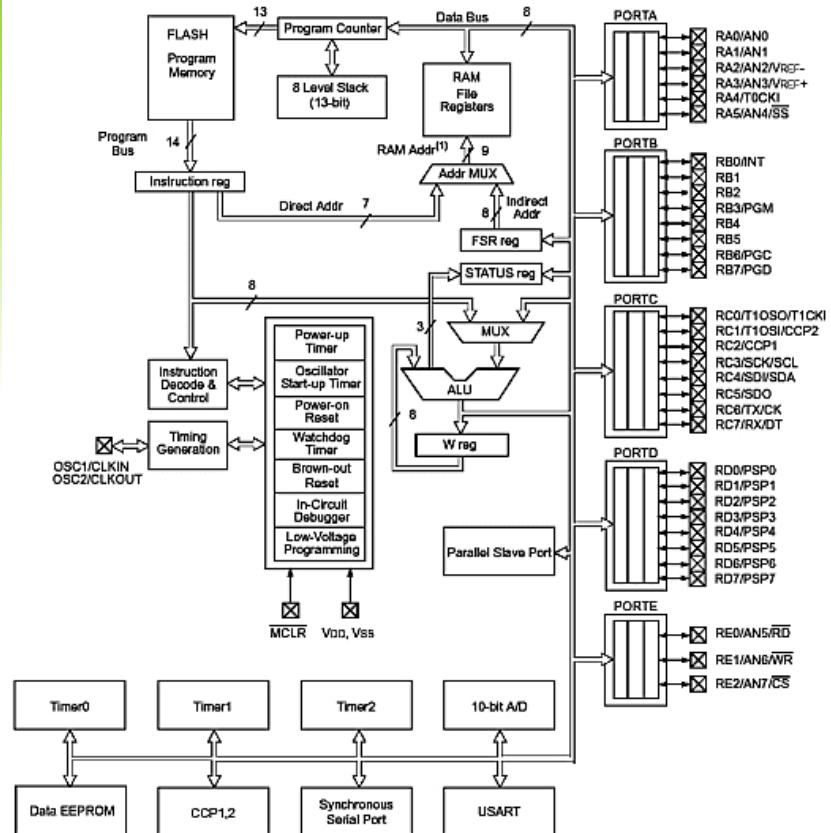


Figura 1.5. Computadora tipo Harvard.

Arquitectura PIC:

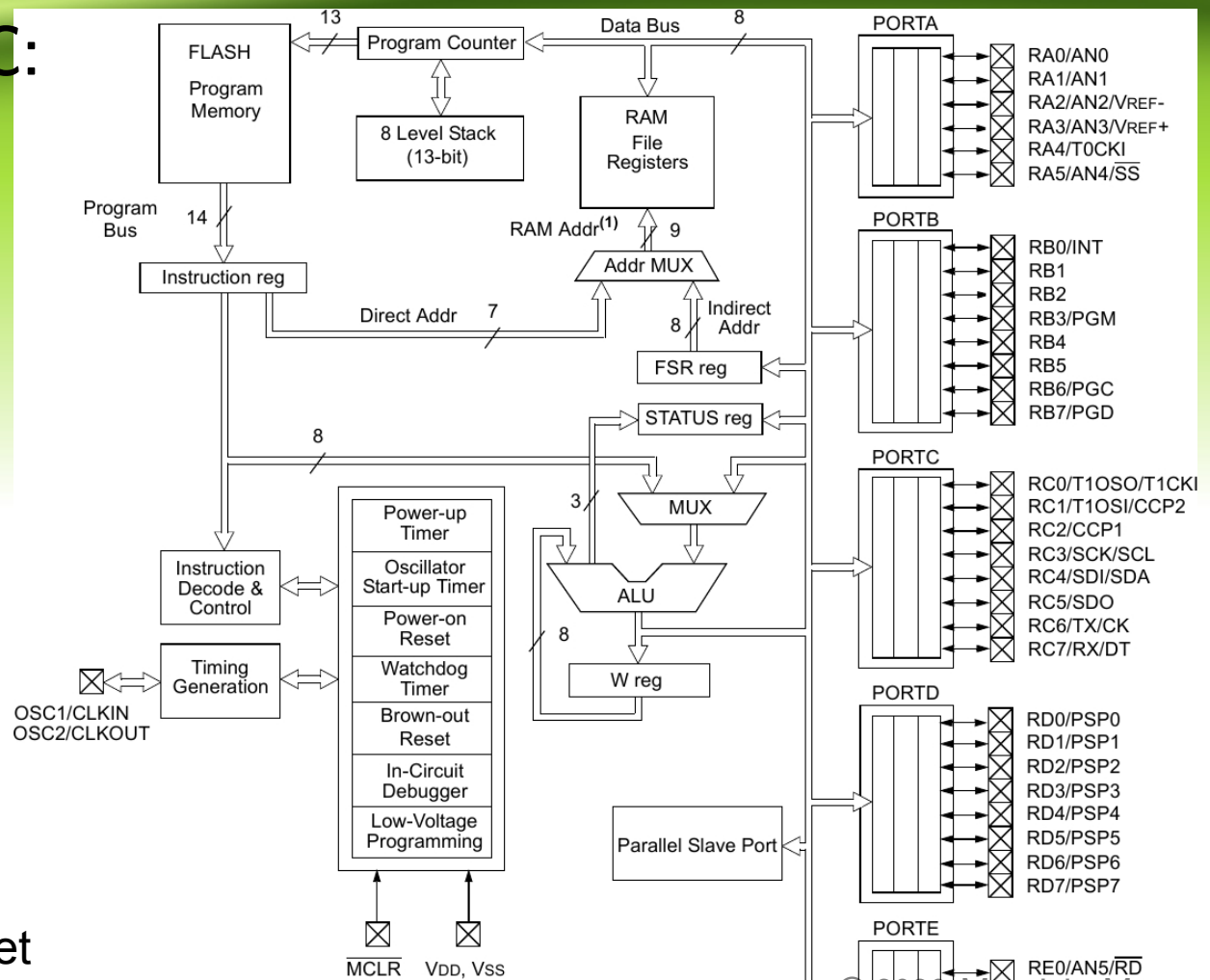
Device	Program FLASH	Data Memory	Data EEPROM
PIC16F874	4K	192 Bytes	128 Bytes
PIC16F877	8K	368 Bytes	256 Bytes



Note 1: Higher order bits are from the STATUS register.

Fuente: 16F874 datasheet

Arquitectura PIC:



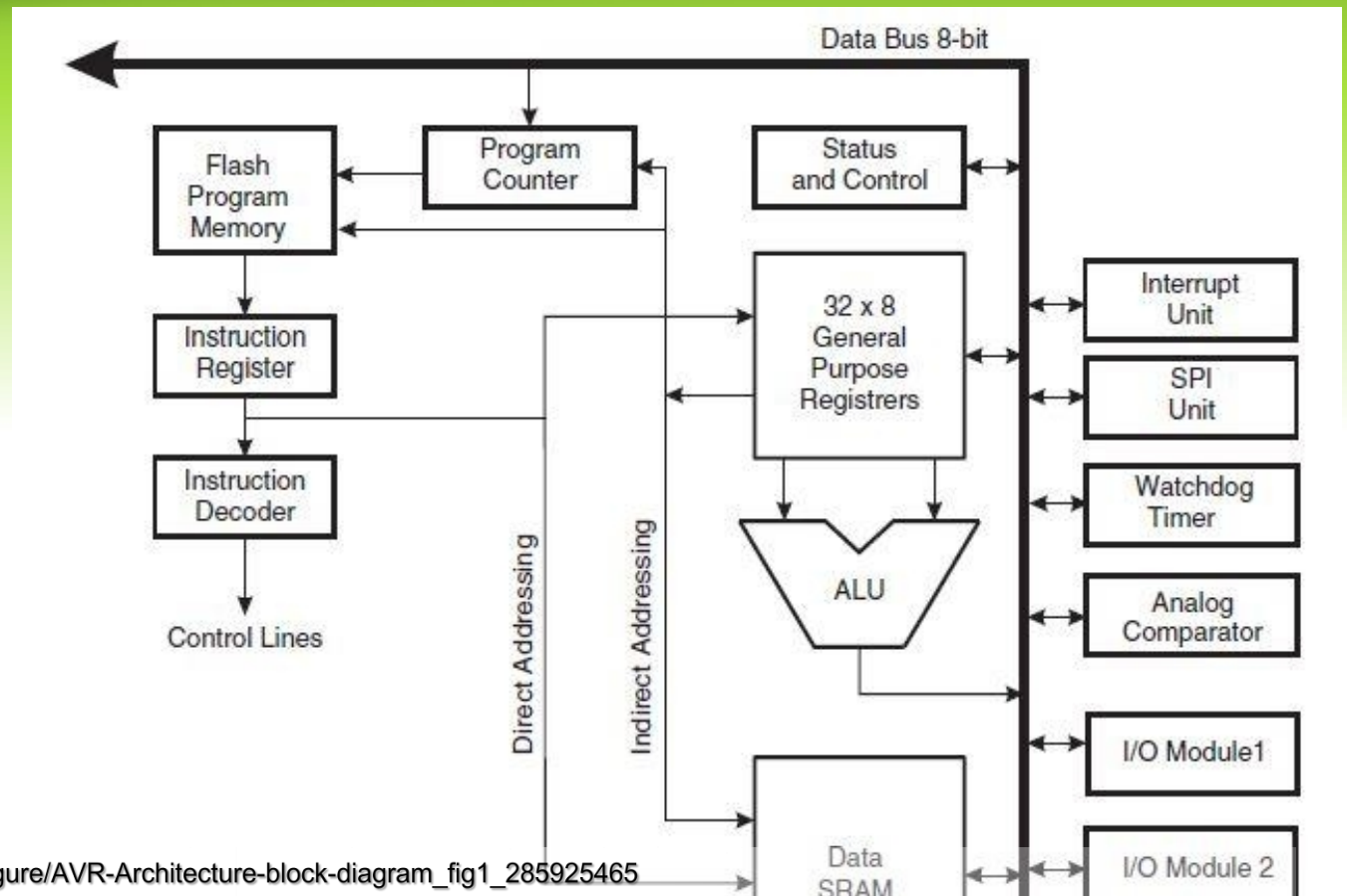
Fuente: 16F874 datasheet

The diagram illustrates the internal architecture of a microcontroller, centered around an 8-bit Data Bus. The components and their connections are as follows:

- Data Bus 8-bit:** A horizontal line at the top with arrows at both ends, representing the system's data bus.
- Program Counter:** Connected to the Data Bus and bidirectionally connected to the Flash Program Memory and Status and Control.
- Flash Program Memory:** Connected to the Program Counter and the Instruction Register.
- Instruction Register:** Receives data from the Flash Program Memory and sends it to the Instruction Decoder.
- Instruction Decoder:** Outputs Control Lines.
- Status and Control:** Connected to the Data Bus and the Program Counter.
- 32 x 8 General Purpose Registers:** Connected to the Data Bus and the ALU.
- ALU (Arithmetic Logic Unit):** Receives data from the General Purpose Registers and outputs to the Data Bus.
- Data SRAM:** Connected to the Data Bus and supports both Direct Addressing (from the Instruction Register) and Indirect Addressing (from the General Purpose Registers).
- EEPROM:** Connected to the Data Bus.
- I/O Lines:** Connected to the Data Bus.
- Peripheral Modules:** All connected to the Data Bus:
 - Interrupt Unit
 - SPI Unit
 - Watchdog Timer
 - Analog Comparator
 - I/O Module 1
 - I/O Module 2
 - I/O Module n

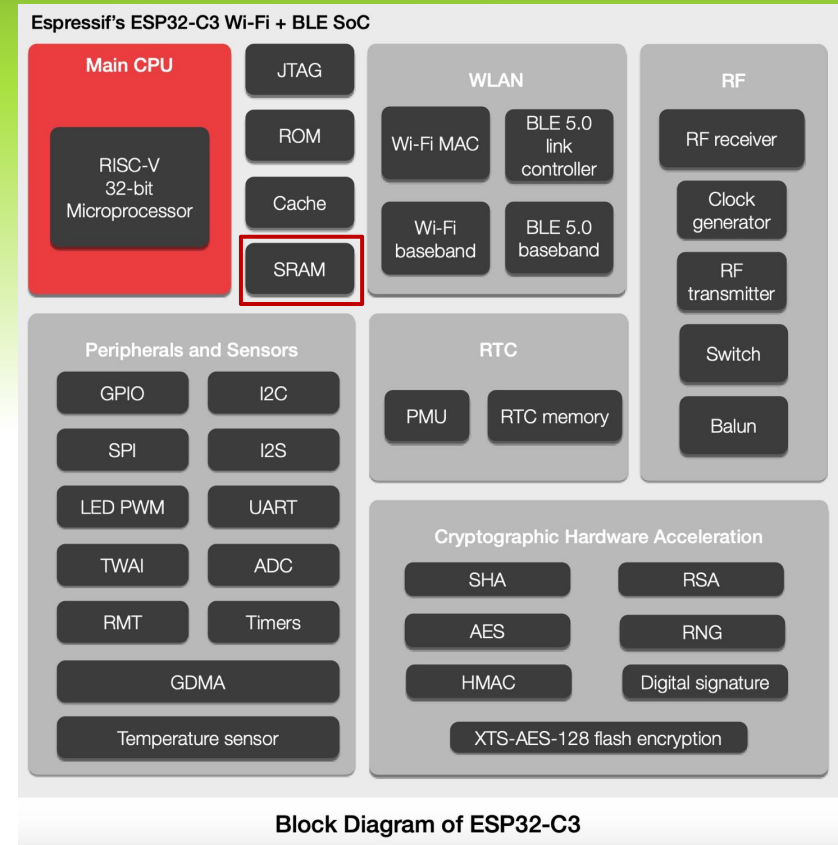
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Arquitectura AVR (Arduino):



Fuente: https://www.researchgate.net/figure/AVR-Architecture-block-diagram_fig1_285925465

Arquitectura Espressif ESP32



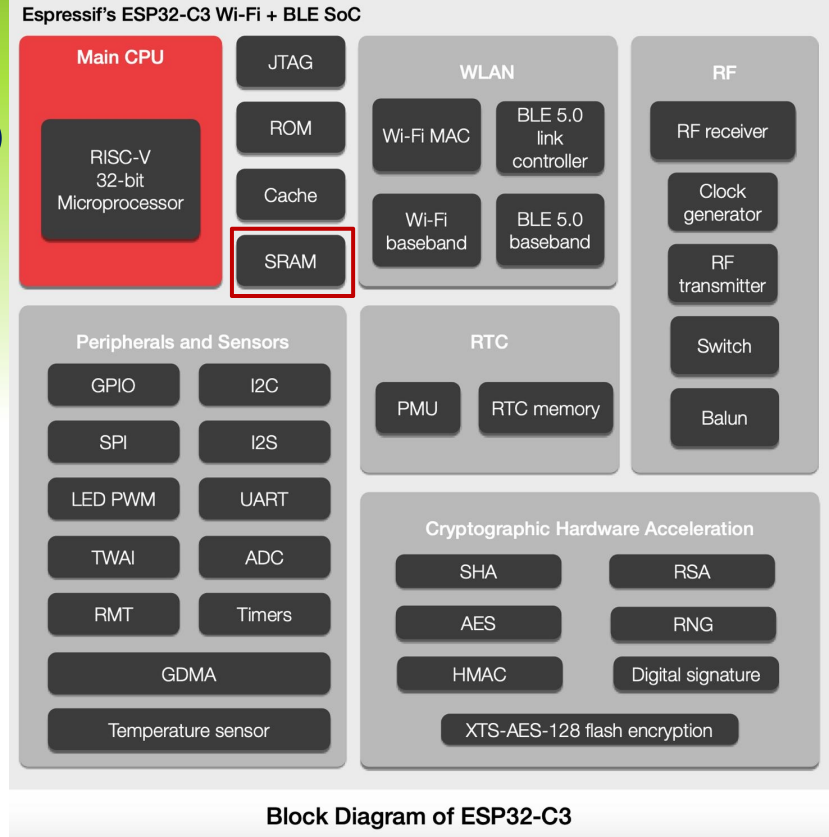
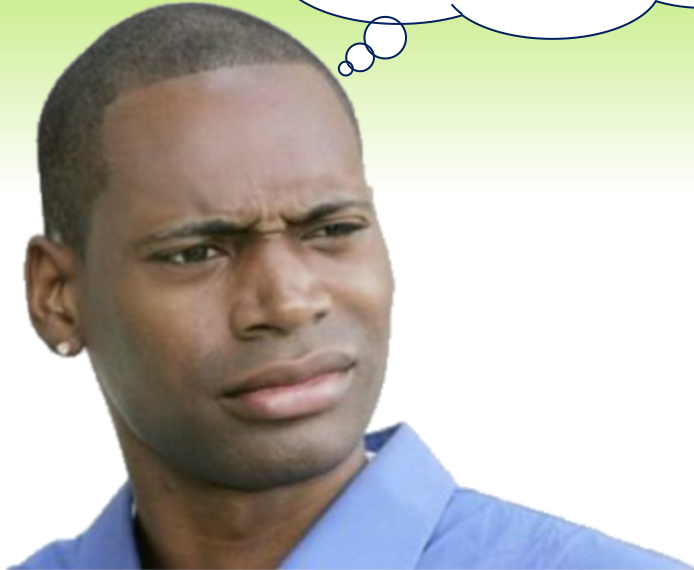
Fuente: https://www.espressif.com/en/news/ESP32_C3

hold up



Arquitectura Espressif ESP32

¿RISC no separa
datos y programa?



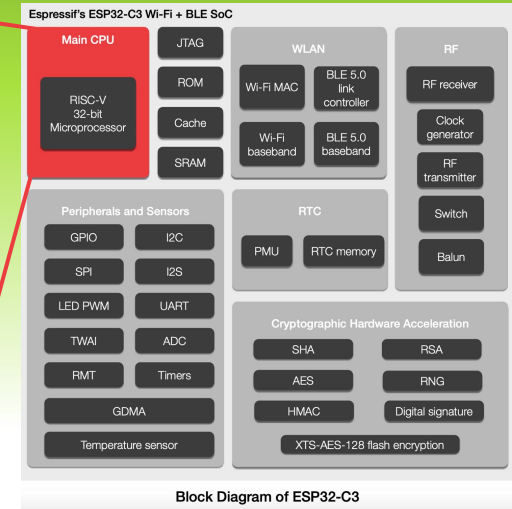
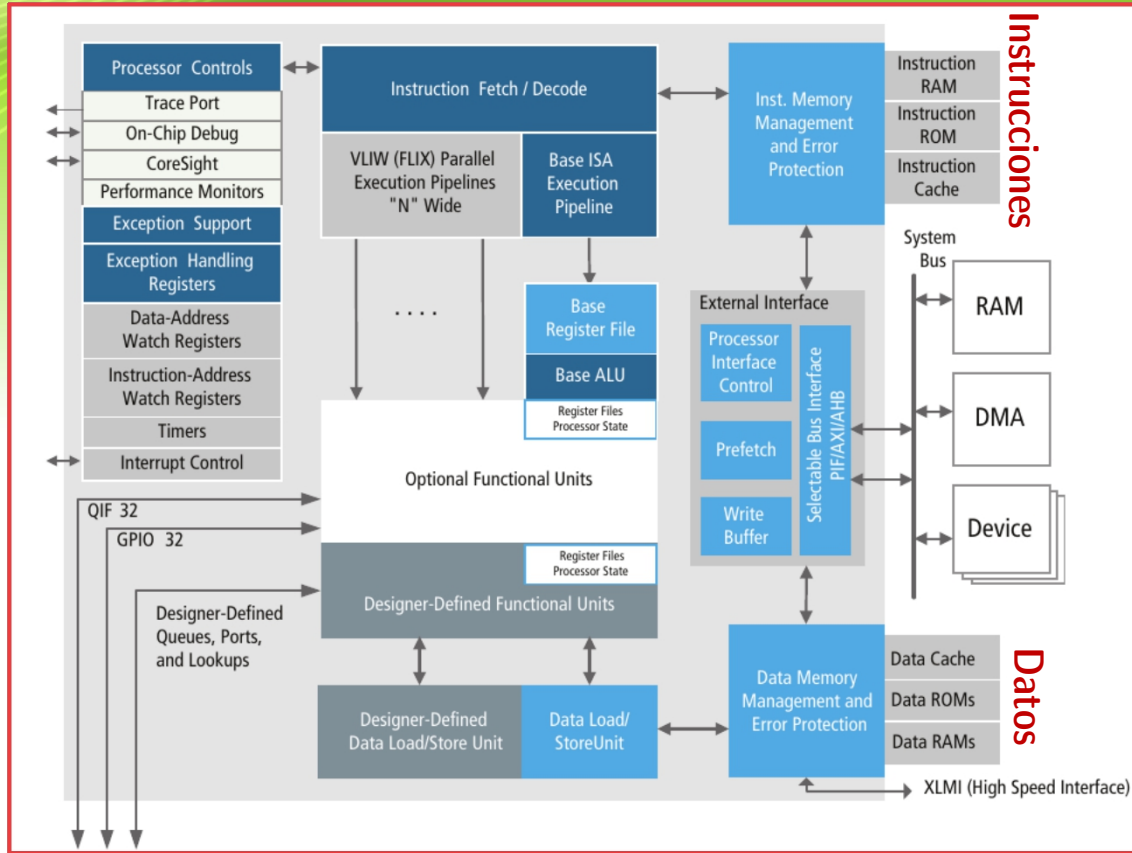
Fuente: https://www.espressif.com/en/news/ESP32_C3

WELL, YEAH...



ACTUALLY...

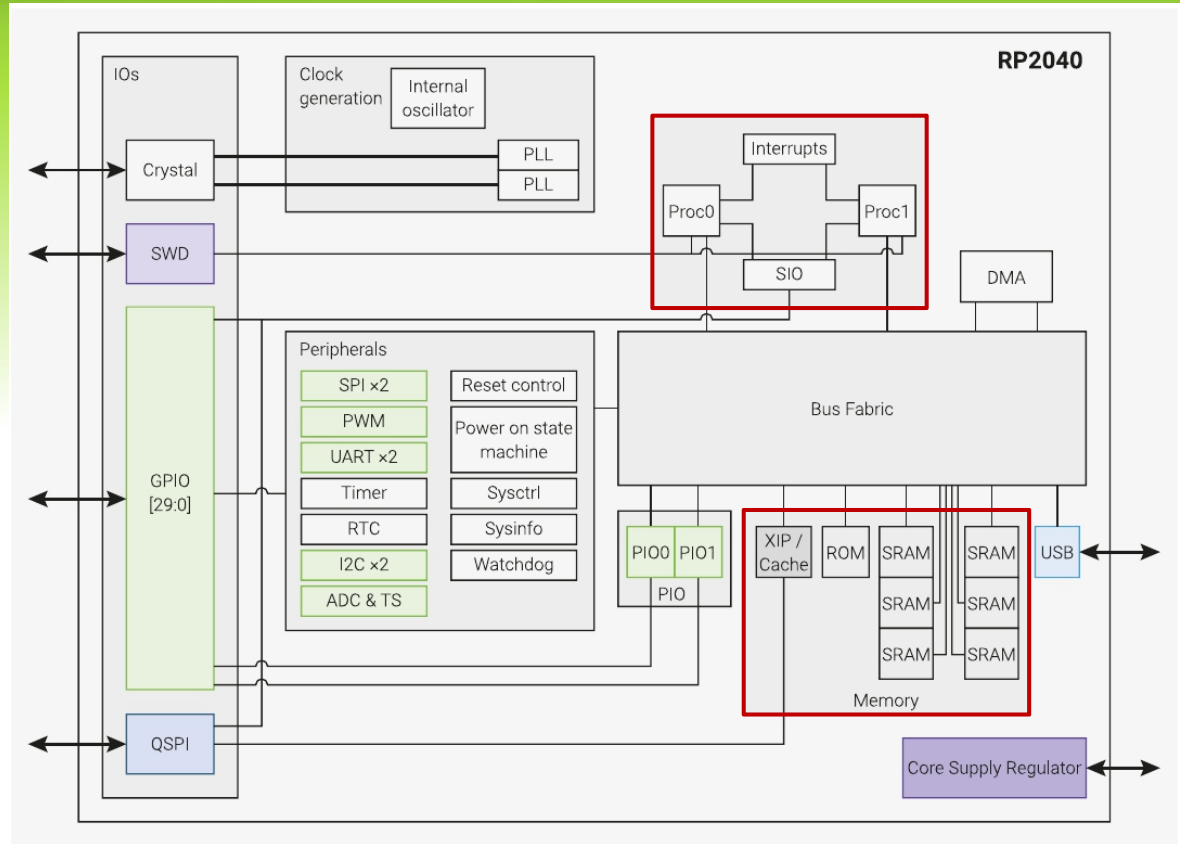
Arquitectura Xtensa LX6 32-bit microprocessor



Fuente: Tensilica Datasheet - Xtensa LX6 Customizable DPU , pp4.

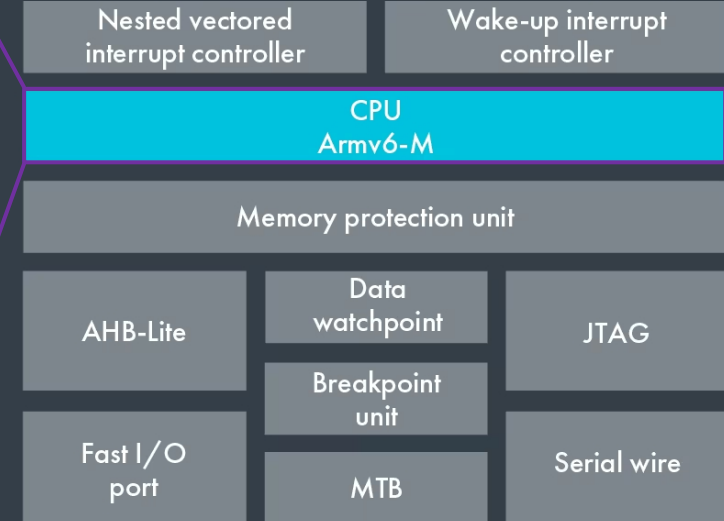
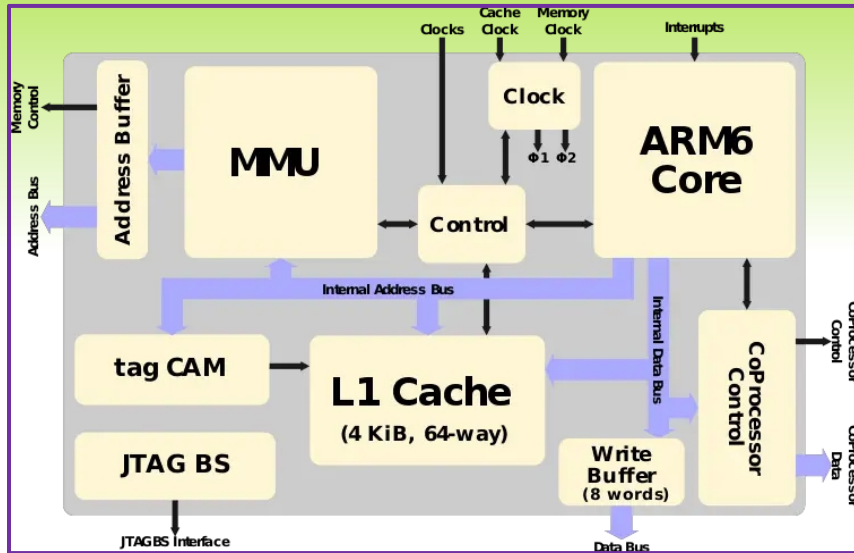


Arquitectura RP2040 (Raspberry Pi Pico)



Fuente: RP2040 Datasheet, pp11. Build version: ae3b121-clean

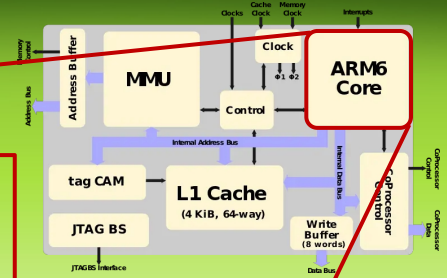
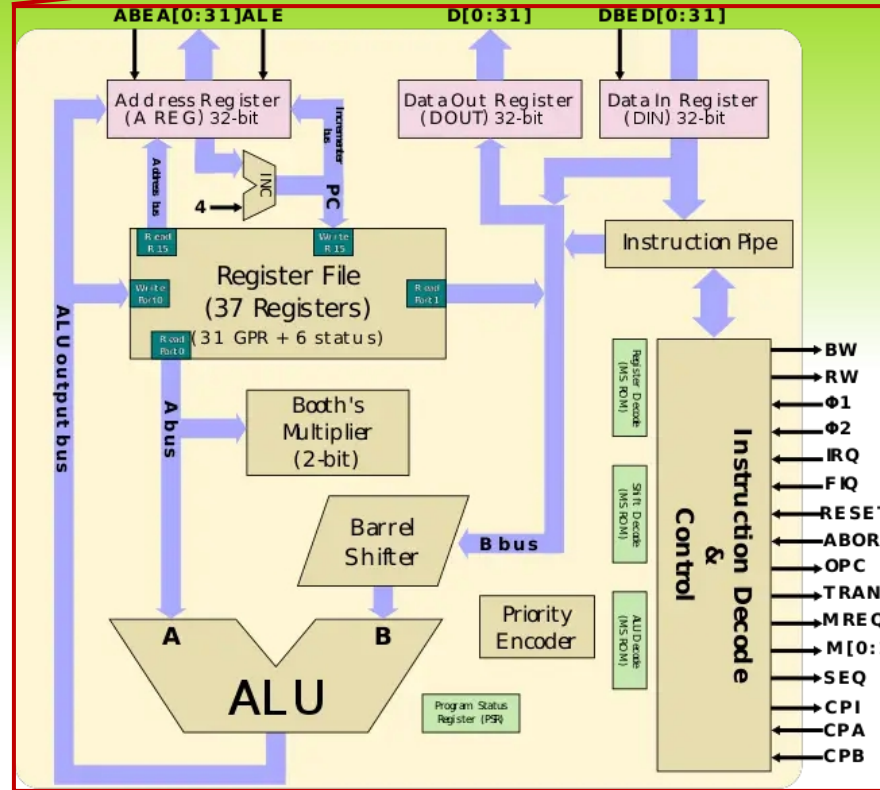
Arquitectura ARM v6-M



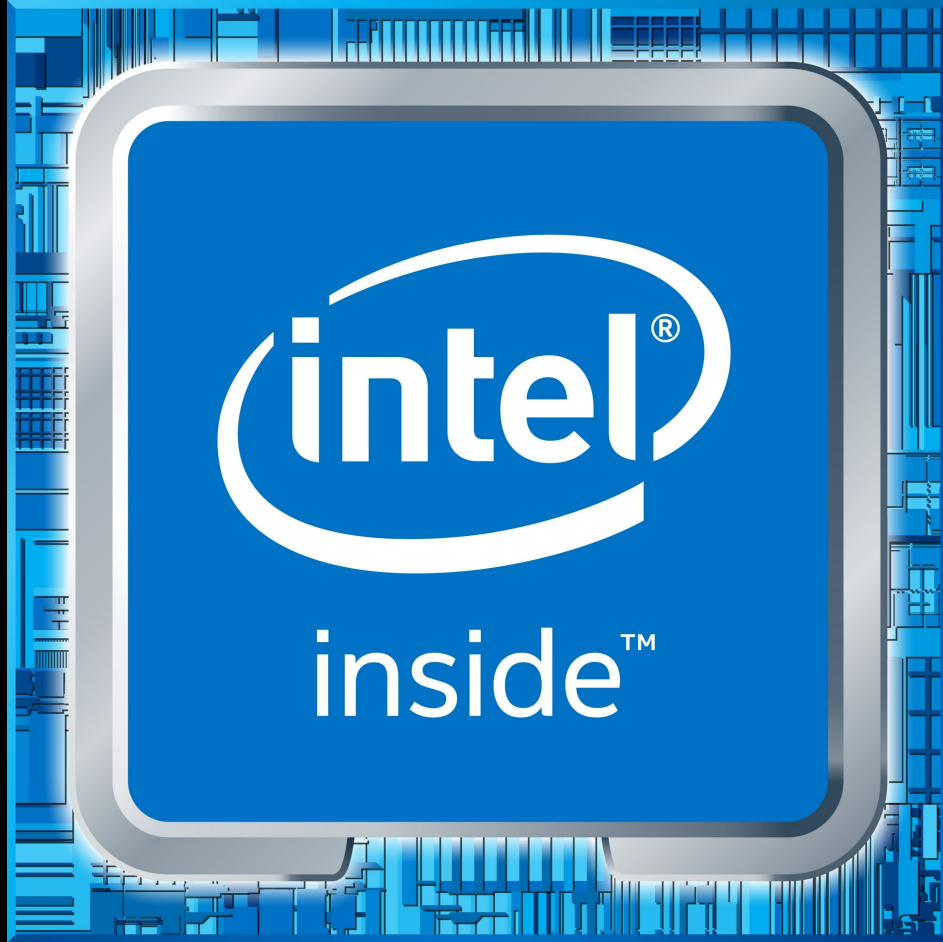
Fuente: <https://developer.arm.com/Processors/Cortex-M0-Plus>

https://en.wikichip.org/wiki/arm_holdings/microarchitectures/arm6

Arquitectura ARM v6-M

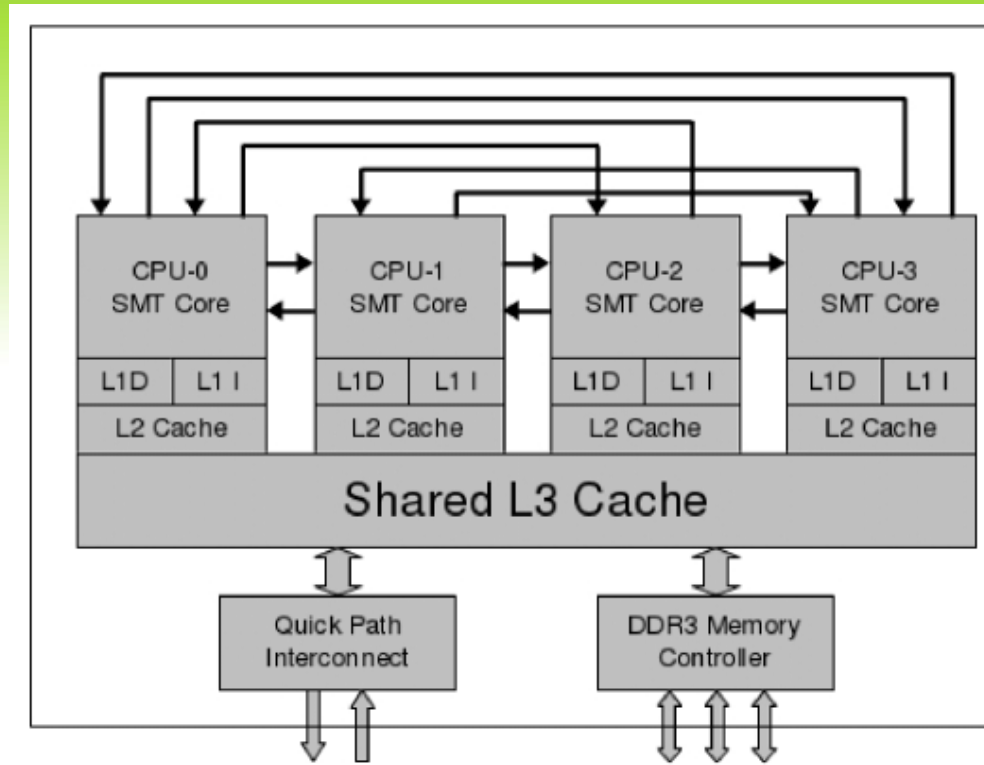


Fuente: https://en.wikichip.org/wiki/arm_holdings/microarchitectures/arm6



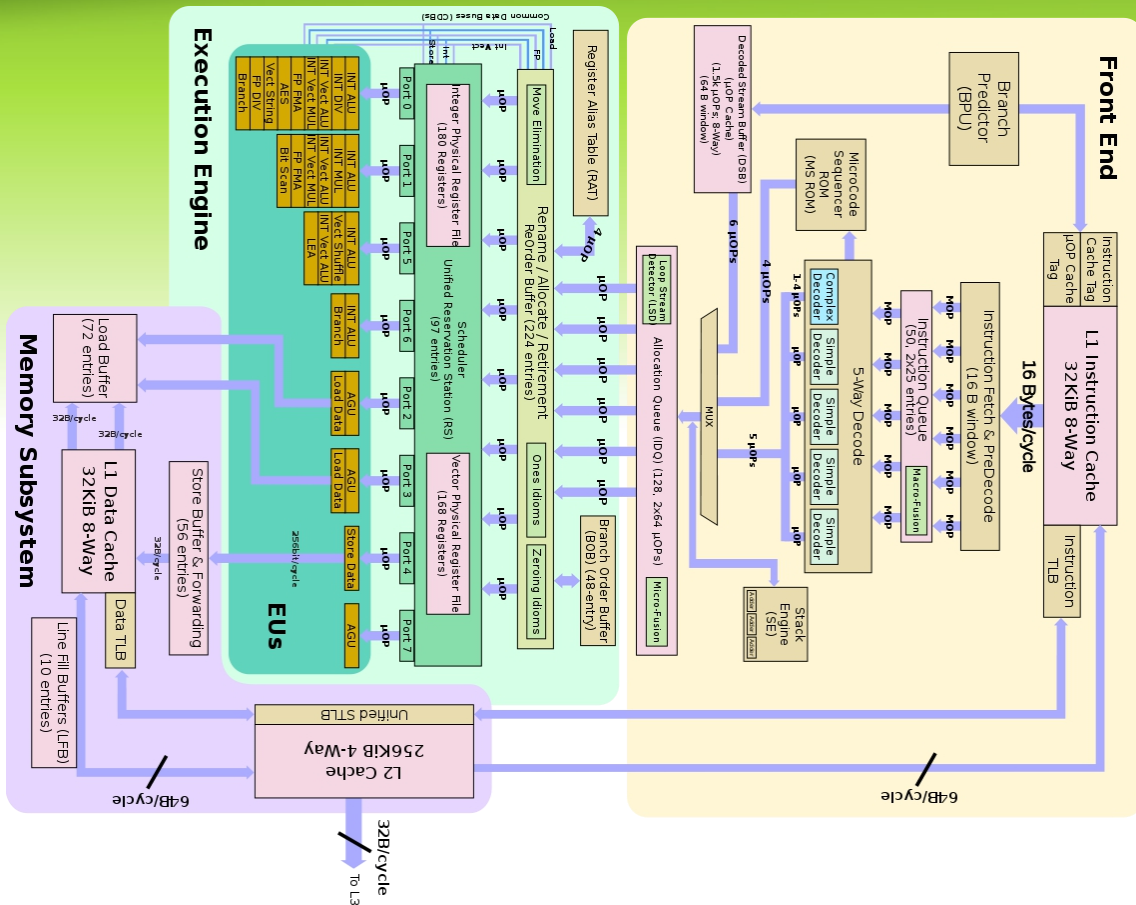
Idiot outside

Arquitectura Intel Core (Core i7):

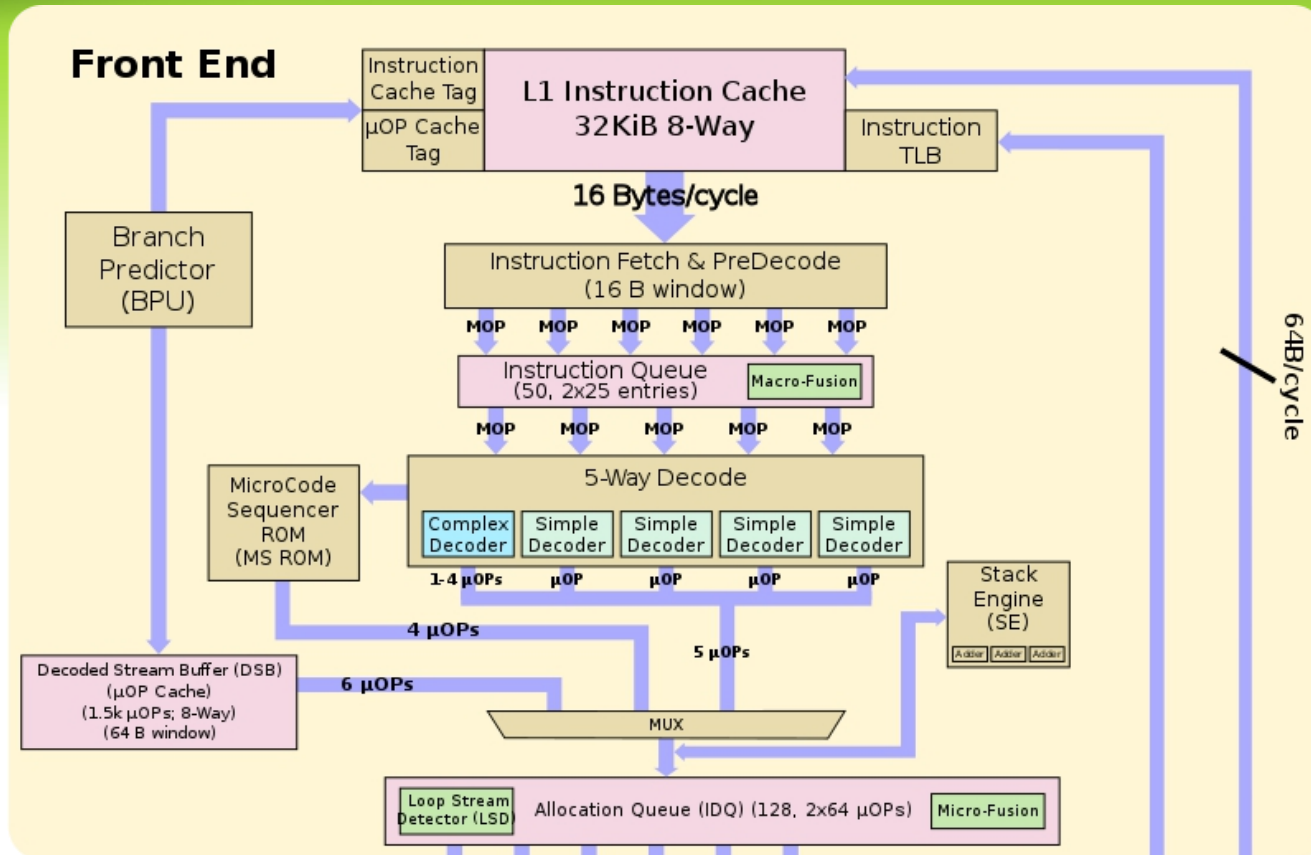


https://www.researchgate.net/figure/Intel-Core-i7-Processor_fig3_267766795

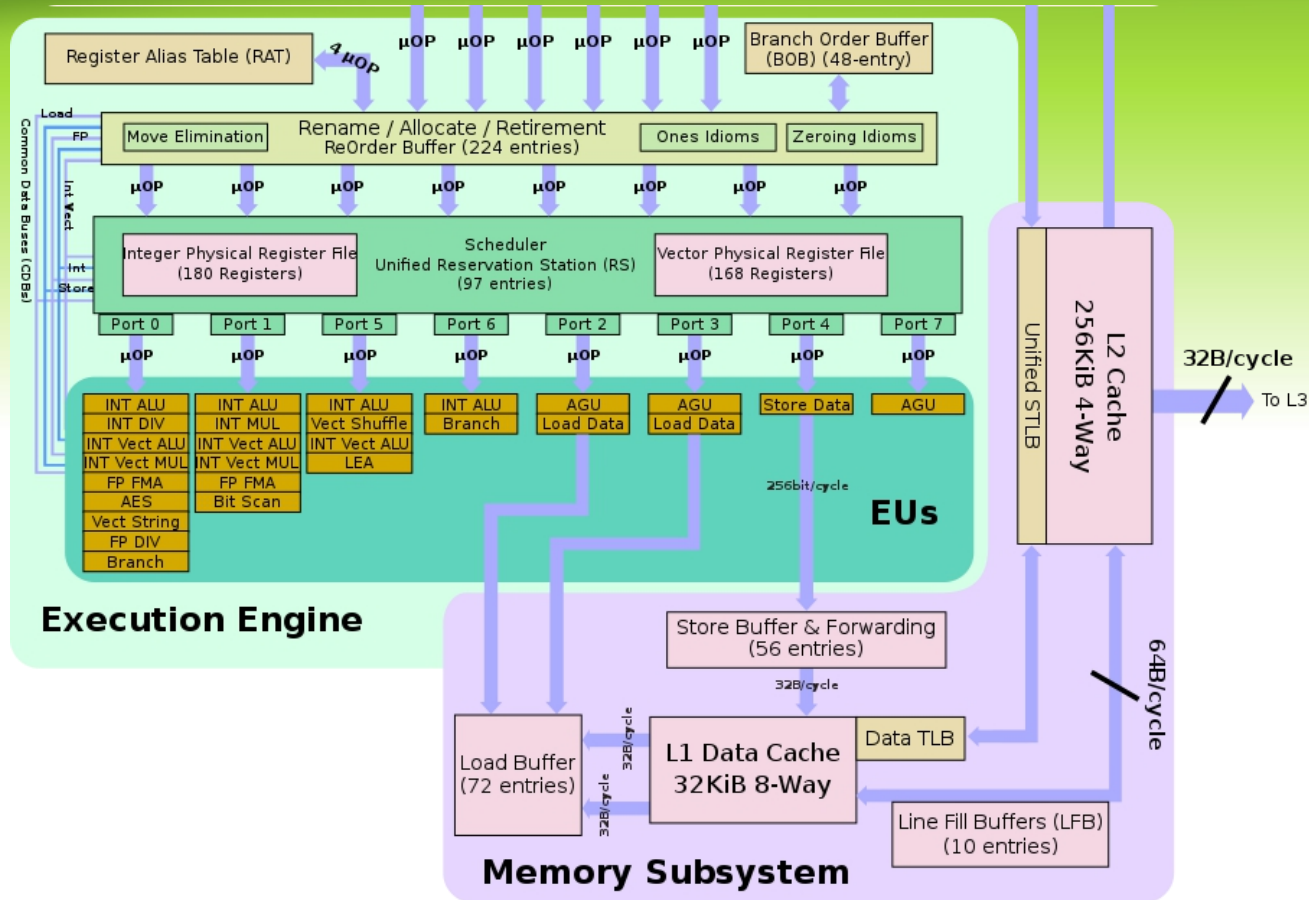
Arquitectura Intel Core (Core i7): Coffee Lake (2017)



Arquitectura Intel Core (Core i7): Coffee Lake (2017)



Arquitectura Intel Core (Core i7): Coffee Lake (2017)



Arquitectura Intel Core: Golden Cove (2021)

